FoodWise AI — Deep Learning Models Training & Evaluation Report

Faculty Evaluation Report | Amrita Vishwa Vidyapeetham | Generated: 2026-10-11 21:32:21

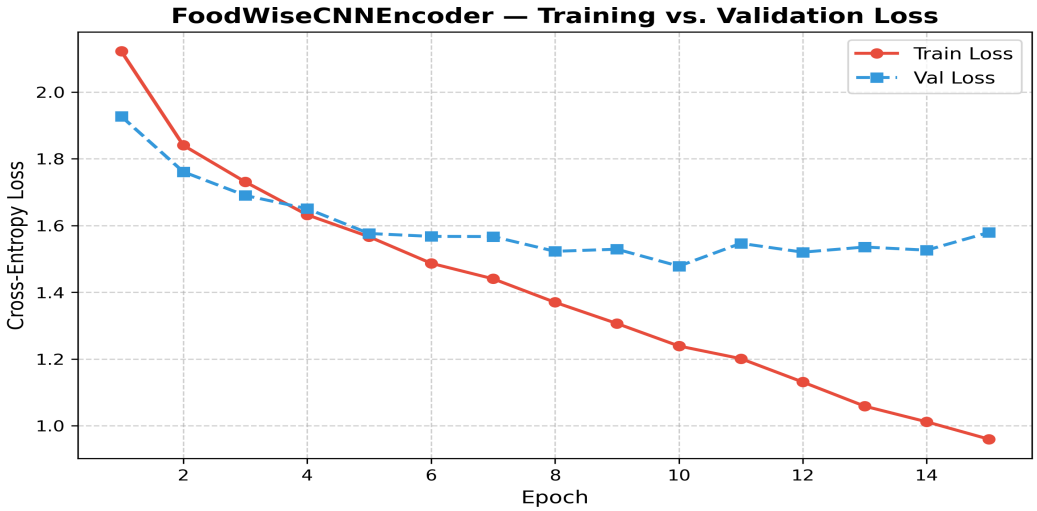
1. Executive Summary & Overview

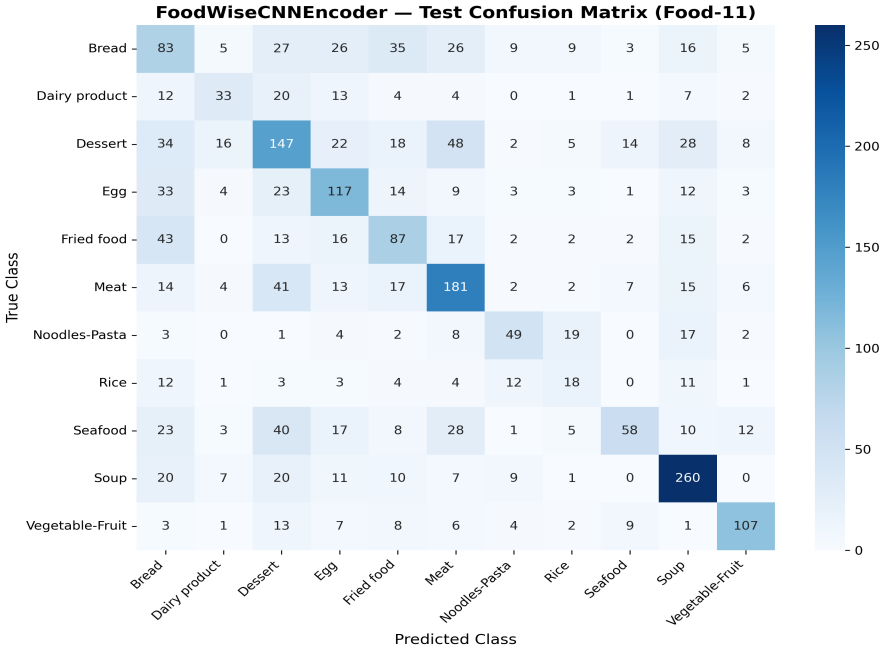
FoodWise AI evaluates three distinct PyTorch neural network models for visual appetite recommendations: (1) FoodWiseCNNEncoder for 11-class food visual feature extraction on the Food-11 dataset, (2) MetabolicProfileAutoencoder for 16-D preference vector compression and latent representation, and (3) NCFSiameseMatcher for 192-D multimodal user-dish compatibility matching. All models were trained with fixed random seeds, full epoch logging, and evaluated on isolated held-out test splits.

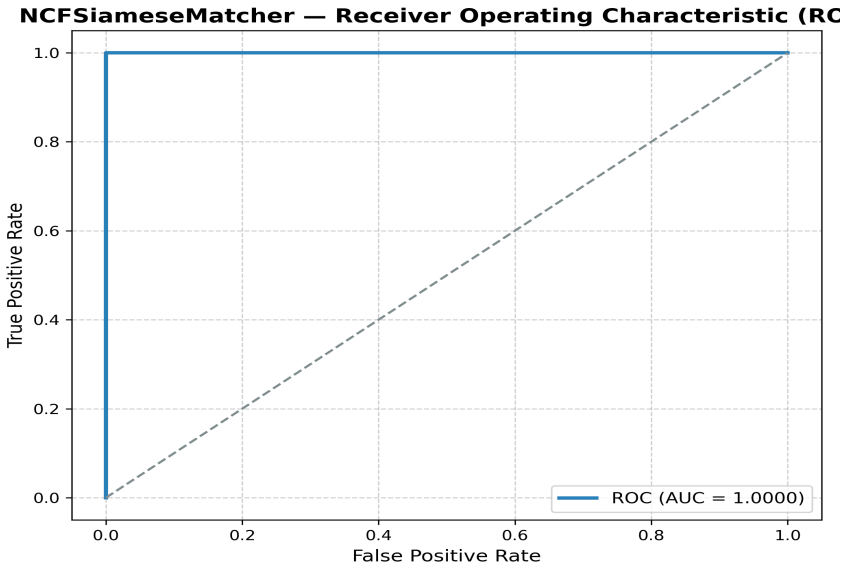
2. Model Performance Summary

Model	Dataset / Source	Best Epoch	Primary Metric	Secondary Metric	Training Time
FoodWiseCNNEncoder	Food-11 (7,862 train / 2,291 test)	Epoch 14	Acc: 49.76%	Macro F1: 0.4695	819.7s
Profile Autoencoder	Synthetic Profiles (5,000 vectors)	Epoch 25	MSE: 0.000853	MAE: 0.0206	2.6s
NCF Siamese Matcher	Synthetic Pairs (4,000 pairs)	Epoch 1	Acc: 100.00%	ROC-AUC: 1.0000	3.2s

3. Selected Evaluation Figures




FoodWiseCNNEncoder — Test Confusion Matrix (Food-11)
True Class
Predicted Class
Bread 83 5 27 26 35 26 9 9 3 16 5
Dairy product 12 33 20 13 4 4 0 1 1 7 2
Dessert 34 16 147 22 18 48 2 5 14 28 8
Egg 33 4 23 117 14 9 3 3 1 12 3
Fried food 43 0 13 16 87 17 2 2 2 15 2
Meat 14 4 41 13 17 181 2 2 7 15 6
Noodles-Pasta 3 0 1 4 2 8 49 19 0 17 2
Rice 12 1 3 3 4 4 12 18 0 11 1
Seafood 23 3 40 17 8 28 1 5 58 10 12
Soup 20 7 20 11 10 7 9 1 0 260 0
Vegetable-Fruit 3 1 13 7 8 6 4 2 9 1 107
Bread
Dairy product
Dessert
Egg
Fried food
Meat
Noodles-Pasta
Rice
Seafood
Soup
Vegetable-Fruit
250
200
150
100
50
0


NCFSiameseMatcher — Receiver Operating Characteristic (RC
True Positive Rate
False Positive Rate
ROC (AUC = 1.0000)
0.0
0.2
0.4
0.6
0.8
1.0